

NAME: _____

PERIOD: _____

A Percent Needs Its Base

AP Statistics · Unit 1 · Topic 1.3

[STAR] THE PROBLEM

Two clubs asked the student council for money to run a Saturday session. Each club surveyed *all* of its members: “Would you attend a Saturday session?” The table shows every response as a count.

At the council meeting, the **Robotics club** presented a **frequency table** (counts) and the **Chess club** presented a **relative-frequency table** (percents). Each club chose the table that made its case look best.

“Would you attend a Saturday session?” (every member surveyed; counts)

Club	Yes	No	Total
Chess	15	3	18
Robotics	27	18	45

FIRST TAKE — one sentence before you work the parts

Which club has more interest in a Saturday session — Chess or Robotics? One sentence and one number.

[BOOK] TABLES FOR A CATEGORICAL VARIABLE (keep on the page)

A **frequency table** gives the count in each category. A **relative-frequency table** gives its proportion or percent: $\text{relative frequency} = \frac{\text{category count}}{\text{group total}}$. Multiply by 100 for a percent. Counts answer ‘how many?’; percents answer ‘what share?’. Always identify the group that a percent describes.

Work (a)–(c).

DOK 1 (a) How many Robotics members said Yes?

DOK 2 (b) Find the percent saying Yes in each club (nearest whole percent). Which is higher?

DOK 3 (c) Which club has more student interest? State whether you mean more people or a larger share, and use the matching numbers. Explain why the other measure favors the other club and what a reader needs to know to compare them fairly.

Frame: By more interest, I mean _____, so I choose _____:
_____.

Frame: The other measure favors _____ because the club totals are _____.
Readers also need _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored **E / P / I**.